

Hitaesh Saravanarajan

Chester Springs, PA | 484-319-7876 | hitaesh.sara@gmail.com | [LinkedIn](#) | [Github](#)

Education

Purdue University

West Lafayette, IN

B.S. Computer Engineering, GPA: 3.83/4.00

Expected May 2027

- Relevant Coursework: Operating Systems, Computer Architecture, Object Oriented Programming, Data Structures and Algorithms, Advanced C Programming, Microprocessor Systems, Python for Data Science

Experience

Vanguard

May 2026 – Aug 2026

Software Engineering Intern

Malvern, PA

- Incoming Software Engineering Intern building internal platforms and tools on the Corporate Systems Team

Kyron Medical

Oct 2025 – Mar 2026

Software Engineer (Part-Time)

Remote

- Built backend scheduling and intake APIs for conversational voice agents handling 100+ daily patient calls, enabling appointment booking outside clinic hours.
- Developed EHR integrations with ModMed and eClinicalWorks using FHIR APIs, PostgreSQL, and AWS services to automate patient intake and scheduling workflows
- Engineered stateful scheduling workflows with fuzzy matching, LLM-powered intake parsing, and fallback systems for reliable appointment routing.

Penske Truck Leasing

Jun 2025 – Aug 2025

Software Engineering Intern

Reading, PA

- Built fullstack fleet inspection platform modernizing quality control workflows across 50K+ trucks, projected to save 6K+ labor hours annually
- Engineered Angular, Spring Boot, and SQL systems enabling dynamically generated inspection workflows, reusable form architectures, and integrated reporting for fleet inspectors
- Collaborated closely with business stakeholders through prototyping, iterative feedback cycles, and Agile development

Purdue University

Nov 2024 – May 2025

Supplemental Instruction Leader - Advanced C Programming

West Lafayette, IN

- Led technical review sessions for 50+ students covering pointers, memory management, file I/O, and core data structure concepts

Projects

Dual-Core RISC-V Processor | *SystemVerilog, FPGA* | [Github](#)

2026

- Designed dual-core pipelined RISC-V processor with caches, forwarding, hazard detection, and branch control logic
- Passed 95%+ validation coverage and achieved 71 MHz synthesized clock frequency through FPGA toolchain testing

TickDB | *C++, Storage Engine* | [Github](#)

2026

- Building append-only time-series database optimized for low-latency tick ingestion and high-speed range queries
- Implementing WAL, checksums, segmented storage, and indexed retrieval paths for durability and fast reads

Volos | *C++, Python, FastAPI* | [Github](#)

2025

- Built low-latency options Greeks engine achieving 11μs mean hot-path latency for repeated pricing workloads
- Exposed native C++ engine through Python bindings and FastAPI for analytics and real-time pricing workflows

Findify | *Python, FastAPI, React* | [Github](#)

2025

- Built internship discovery platform aggregating listings from 100+ company career sites per hour
- Designed searchable dashboard enabling centralized internship discovery and reducing manual search time

Technical Skills

Languages: C, C++, Python, Java, TypeScript, SQL, SystemVerilog, RISC-V Assembly

Systems / Backend: Spring Boot, FastAPI, Flask, PostgreSQL, MySQL, REST APIs

Frontend: React, Angular, React Native, Tailwind CSS

Tools / Cloud: Git, AWS, GCP, Linux, Docker, Selenium, BeautifulSoup